

Cristian Diaz

Computer Science Engineer | Frontend Developer & QA Automation

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<https://github.com/cristian-diaz2402> - Portfolio: <https://cristian-diaz2402.github.io/CristianDiazPortafolio/>

Professional Profile

Computer Science Engineer (EPN, 2026) with over 2 years of production-level experience. I specialize in developing frontend modules using Vue.js and React, and I lead end-to-end (E2E) testing architectures with Playwright on GitHub Actions CI/CD pipelines. My expertise includes designing event-driven systems on AWS and engineering a local Retrieval-Augmented Generation (RAG) architecture capable of processing 26,000 concurrent queries with a 23ms latency. I am seeking a Frontend or QA Automation role where quality is integrated from the very beginning of the development lifecycle.

Languages: Spanish (Native) - English (**B1 / B2**)

Professional Experience

Frontend Developer & QA Automation | Wanqara – E-Invoicing SaaS for PYMES

Jun 2024 – Aug 2026 - Contract - Remote - Ecuador

- Developed and actively maintain two client-facing frontend modules currently in production: a Kitchen Display System (KDS) for real-time order management and a POS Status Viewer, leveraging Vue.js, Pinia, and TanStack Query.
- Designed and implemented the E2E testing suite from scratch using Playwright (300+ active tests), integrating smoke tests in the develop branch and comprehensive regression testing upon merging to main.
- Configured CI/CD pipelines in GitHub Actions featuring concurrency management and adaptive API_TIMEOUTs, successfully resolving race conditions in waitForResponse that previously caused false negatives during CI runs.

Frontend Developer | AntsOfEcuador – Scientific Biodiversity Tracking Platform

May 2025 – Jul 2025 - Freelance - Production-Ready - Ecuador

- Engineered and deployed a scientific platform for tracking ant species (React + FastAPI + PostgreSQL) within a 2-month timeframe, effectively replacing manual spreadsheet-based workflows.
- Built a data import pipeline with automated taxonomic validation against the AntCat API, integrated with D3.js geospatial visualizations for regional analysis.

Featured Projects

3D Virtual Assistant with RAG Architecture – Capstone, EPN 2026

- Implemented local RAG using Qdrant as an alternative to model fine-tuning; this architecture allows for straightforward embedding updates compared to the costly retraining required when institutional regulations frequently change.
- Utilized vLLM (PagedAttention) paired with Redis semantic caching for inference, successfully processing 26,000 concurrent queries with zero failures, 23ms latency, and 100% prompt injection mitigation. Developed a 3D frontend using Vue 3 and Three.js, deployed on a Linux/Nginx server with SSL.

Stack: *Vue 3 - Three.js - FastAPI - RAG - Qdrant - vLLM - Redis - Python - Nginx*

Shoplight — E-commerce Cloud & EDA - <https://github.com/cristian-diaz2402/ShoplightAWS>

- Designed an Event-Driven Architecture (EDA) on AWS: configured SQS as a buffer for order spikes, SNS for inventory alerts, Multi-AZ PostgreSQL within a private VPC, DynamoDB for immutable auditing, and Rekognition for automated visual tagging.
- Validated system resilience through Chaos Engineering on Elastic Beanstalk, achieving automatic recovery with zero service interruption.

Stack: *Vue 3 - Node.js - Docker - AWS (VPC, EC2, RDS, S3, SQS, SNS, DynamoDB, Rekognition)*

Capibara Bot — Robotics, HCI & ML - <https://github.com/cristian-diaz2402/CapibaraBot>

- Built an ESP32-based educational robot controlled via an Android Bluetooth app. Applied a Random Forest model for user telemetry (using a low-dimensional tabular dataset for an interpretable and lightweight model). Evaluated usability using the SUS scale, achieving results comparable to industry benchmarks.

Stack: *ESP32 - C++ - Android (Kotlin) - Python - FastAPI - Random Forest (Scikit-learn)*

Education

B.S. in Computer Science Engineering - Escuela Politécnica Nacional (EPN), Quito, Ecuador
- Graduated August 2026

Technical Skills

Languages & Frameworks: *Vue.js - React - TypeScript - Python - FastAPI - Node.js*

Infrastructure & Databases: *AWS (EC2, S3, RDS, VPC, SQS, SNS) – Docker - PostgreSQL - Qdrant - Redis - DynamoDB - Nginx*

Quality Assurance & Tools: *Playwright - GitHub Actions (CI/CD) - Git - Jira - Linux (Ubuntu)*

AI & Machine Learning: *vLLM - Ollama - LangChain - RAG - Random Forest (Scikit-learn)*

Hardware & Systems: *Embedded C++ (ESP32) - OpenGL / GLSL – Distributed Systems (Oracle).*

Agile Methodology: *Weekly sprints, weekly production release, Kanban board.*